PSEUDO-COHERENCE ASCENDS ALONG A FINITE ORDER THICKENING OF SCHEMES

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 20743-20793, immediately following the statement at 20736-20741.

Editorial note. Seven cross-chapter pointers (Cohomology, More on Algebra, Derived Categories of Schemes) plus the chapter's own thickenings section are recorded in assembly.external_references.

Editorial note. Same-name pair note: the spaces side of this label delegates its core case to this proof, so the pair is counted once, here.

Allowed standard prerequisites (editorial).

- Section on thickenings. chapter Section section-thickenings, cited at proof line 20746. Reduces the general finite order thickening to a first order one.
- Pseudo-coherence modulo a nilpotent. More on Algebra, lemma more-algebra-lemma-check-pseudo-coherent-modulo-nilpotent, cited in the proof. Is the algebraic criterion that lets pseudo-coherence be tested after reduction modulo the nilpotent ideal.
- Affine comparison for bounded complexes. Derived Categories of Schemes, lemma perfect-lemma-affine-compare-bounded, cited at proof lines 20784-20785. Represents the object by a complex of modules over A' so that the ring-theoretic criterion applies.
- Quasi-coherence of direct images and tensor products. Derived Categories of Schemes, lemmas perfect-lemma-quasi-coherence-direct-image and perfect-lemma-quasi-coherence-tensor-product, cited in the proof. Establishes that K' has quasi-coherent cohomology sheaves, the first half of the proof.
- Projection formula for a closed immersion. Cohomology, lemma cohomology-lemma-projection-formula-closed-immersion, cited in the proof. Computes the restriction of K' to X in the short exact sequence of the thickening.

Source reference. Original reference 0B55: focused-seeds/geometry/S9-morph-thickening-pseudo-coherent-schemes/ref-tags.txt, source lines 4–4. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0H76: focused-seeds/geometry/S9-morph-thickening-pseudo-coherent-schemes/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 06Z0: focused-seeds/geometry/S9-morph-thickening-pseudo-coherent-schemes/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction

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the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08E5: focused-seeds/geometry/S9-morph-thickening-pseudo-coherent-schemes/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08E7: focused-seeds/geometry/S9-morph-thickening-pseudo-coherent-schemes/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08D5: focused-seeds/geometry/S9-morph-thickening-pseudo-coherent-schemes/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08DX: focused-seeds/geometry/S9-morph-thickening-pseudo-coherent-schemes/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original section 04EW: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 30–30. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $i : X \to X'$ be a finite order thickening of schemes. Let $K' \in D(\mathcal{O}_{X'})$ be an object such that $K = Li^*K'$ is pseudo-coherent. Then K' is pseudo-coherent.*

Proof. We first prove K' has quasi-coherent cohomology sheaves. To do this, we may reduce to the case of a first order thickening, see Section 04EW. Let $\mathcal{I} \subset \mathcal{O}_{X'}$ be the quasi-coherent sheaf of ideals cutting out X. Tensoring the short exact sequence

$$0 \to \mathcal{I} \to \mathcal{O}_{X'} \to i_*\mathcal{O}_X \to 0$$

with K' we obtain a distinguished triangle

$$K' \otimes^{\mathbf{L}}_{\mathcal{O}_{X'}} \mathcal{I} \to K' \to K' \otimes^{\mathbf{L}}_{\mathcal{O}_{X'}} i_*\mathcal{O}_X \to (K' \otimes^{\mathbf{L}}_{\mathcal{O}_{X'}} \mathcal{I})[1]$$

Since $i_* = Ri_*$ and since we may view $\mathcal{I}$ as a quasi-coherent $\mathcal{O}_X$-module (as we have a first order thickening) we may rewrite this as

$$i_*(K \otimes^{\mathbf{L}}_{\mathcal{O}_X} \mathcal{I}) \to K' \to i_*K \to i_*(K \otimes^{\mathbf{L}}_{\mathcal{O}_X} \mathcal{I})[1]$$

Please use Cohomology, Lemma 0B55 to identify the terms. Since K is in $D_{QCoh}(\mathcal{O}_X)$ we conclude that K' is in $D_{QCoh}(\mathcal{O}_{X'})$; this uses Derived Categories of Schemes, Lemmas 08E5, 08DX, and 08D5.

Assume K' is in $D_{QCoh}(\mathcal{O}_{X'})$. The question is local on X' hence we may assume X' is affine. Say $X' = \mathrm{Spec}(A')$ and $X = \mathrm{Spec}(A)$ with $A = A'/I$ and I nilpotent. Then K' comes from an object $M' \in D(A')$, see Derived Categories of Schemes, Lemma 06Z0. Thus $M = M' \otimes^{\mathbf{L}}_{A'} A$ is a pseudo-coherent object of $D(A)$ by Derived

Categories of Schemes, Lemma 08E7 and our assumption on K. It follows that M' is pseudo-coherent by More on Algebra, Lemma 0H76. This finishes the proof. $\square$